# GobletNet: Wavelet-Based High-Frequency Fusion Network for Semantic Segmentation of Electron Microscopy Images

Yanfeng Zhou, Lingrui Li, Chenlong Wang, Le Song, *Member, IEEE*, and Ge Yang, *Member, IEEE*

***Abstract*— Semantic segmentation of electron microscopy (EM) images is crucial for nanoscale analysis. With the development of deep neural networks (DNNs), semantic segmentation of EM images has achieved remarkable success. However, current EM image segmentation models are usually extensions or adaptations of natural or biomedical models. They lack the full exploration and utilization of the intrinsic characteristics of EM images. Furthermore, they are often designed only for several specific segmentation objects and lack versatility. In this study, we quantitatively analyze the characteristics of EM images compared with those of natural and other biomedical images via the wavelet transform. To better utilize these characteristics, we design a high-frequency (HF) fusion network, GobletNet, which outperforms state-of-the-art models by a large margin in the semantic segmentation of EM images. We use the wavelet transform to generate HF images as extra inputs and use an extra encoding branch to extract HF information. Furthermore, we introduce a fusion-attention module (FAM) into GobletNet to facilitate better absorption and fusion of information from raw images and HF images. Extensive benchmarking on seven public EM datasets (EPFL, CREMI, SNEMI3D, UroCell, MitoEM, Nanowire and BetaSeg) demonstrates the effectiveness of our model. The code is available at https://github.com/Yanfeng-Zhou/GobletNet.**

***Index Terms*— Semantic segmentation, electron microscopy images, wavelet, image characteristics, prior knowledge.**

## I. INTRODUCTION

SEMANTIC segmentation is a fundamental task in computer vision, where the goal is to assign a class label to each pixel. Semantic segmentation of natural images has achieved superior results with the development of convolutional neural networks (CNNs) [1], [2], [3], [4]. The use of backbone- and segmentation head-based architectures has become the dominant strategy [5], [6]. Moreover, biomedical image semantic segmentation has advanced, and efficient encoder-decoder architectures have achieved superior performance [7], [8], [9], [10], [11], [12]. Recently, sequence-to-sequence transformers have become popular for semantic segmentation [13], [14]. Several studies have attempted to combine CNNs with transformers [15], [16], enabling the model to take advantage of both the low computational cost of CNNs and the global receptive field of transformers.

Electron microscopy (EM) image semantic segmentation can be regarded as a branch of biomedical imaging, and it is highly important. EM is widely used for imaging at the nanometer scale to study tissues, cells, subcellular structures and molecular complexes [17]. With faster and automated image acquisition, one electron microscope can acquire petabytes of data in a short time [18], [19]. Compared with time-consuming and costly manual annotation, automatic EM semantic segmentation methods are crucial. Several studies have explored specialized models for the semantic segmentation of EM images, such as the residual deconvolutional network (RDN) [20], the deep contextual residual network (DCR) [21], FusionNet [22], and DenseUNet [23].

However, current specialized models for EM images are often extensions or adaptations of natural or biomedical models. They lack the full exploration and utilization of the intrinsic characteristics of EM images. Furthermore, they are often designed only for several specific segmentation objects and lack versatility. For example, [24] proposed a fast mitochondria segmentation method based on modified UNet. Reference [25] proposed a method that combines affinity prediction and region agglomeration for neuron segmentation from EM images, which can reduce topological errors for tubular and network-like segmentation objects but may have little effect on other targets (such as mitochondria and nuclei).

In this study, we quantitatively analyze the characteristics of EM images via the wavelet transform. Compared with those of natural and other biomedical images, the high-frequency (HF) components of EM images have richer texture details and clearer object contours, but also contain more noise. Adding

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low-frequency (LF) components to HF components in an appropriate proportion can reduce noise interference. To better utilize these characteristics, we propose a wavelet-based HF fusion network, GobletNet. GobletNet uses a wavelet transform to generate HF images as extra inputs and uses an extra encoding branch to extract HF information (such as texture and contours) from HF images, which is also equivalent to introducing additional prior knowledge into the model. We introduce a fusion-attention module (FAM) at each stage of the encoding process to facilitate better absorption and fusion of information from raw images and HF images. Our model achieves state-of-the-art performance on seven public datasets with different segmentation objects, data scales, imaging regions and EM types.

*Motivation:* With the emergence of various well-designed segmentation modules, such as dilated convolution [26], feature pyramids [27], and attention mechanisms [28], CNNs are robust and strong enough to extract features from images. To further improve segmentation performance, we should use the characteristics of segmented images to drive the architecture design, so that the architecture can better suit the image characteristics. Based on our previous studies [29], we find that EM images are more sensitive to the wavelet transform than other images are, i.e., HF components acquired by the wavelet transform can highlight texture and contour details but also contain more noise. Texture and contour details are crucial for accurate boundary segmentation. Based on this characteristic, we construct a dual-branch encoder framework to better utilize HF components and suppress noise interference.

Our contributions are summarized as follows:

- We quantitatively analyze the characteristics of EM images via the wavelet transform.
- We propose a wavelet-based HF fusion network, GobletNet, which achieves state-of-the-art semantic segmentation of EM images.
- Extensive benchmarking on seven public datasets demonstrates the effectiveness of our model.

## II. Related Work

### A. Natural and Biomedical Image Semantic Segmentation

Since FCN [1] was proposed as the first fully convolutional neural network for semantic segmentation, backbone- and segmentation head-based architectures have gradually become the mainstream paradigm for natural image semantic segmentation [2], [5]. Studies have proposed various strategies to help the model better extract features, including using dilated convolutions to expand the receptive field [5], [26], using multi-scale fusion to retain spatial and positional details and enhance the global context information [3], [30], and using non-local attention mechanisms to capture long-range dependencies [4], [6]. For biomedical images, efficient and lightweight encoder–decoder architectures, such as UNet [7] and UNet++ [8], are dominant. Compared with the backbone-head architecture, these architectures can retain the position information of boundaries and alleviate overfitting under the limited data volume of medical images. Furthermore, studies have extended these architectures to 3D to meet the needs of volumetric segmentation; for example, [11] proposed a 3D fully CNN VNet, [12] extended UNet to 3D, and ConResNet [31] proposed inter slice context residual learning. Recently, transformers have achieved impressive results in semantic segmentation [13], [14]. Based on the multi-resolution parallel design of HRNet [15], [30] proposed a high-resolution transformer. UNETR [16] uses a transformer as the encoder to learn sequence representations of input images to effectively capture global multi-scale information and uses a CNN as the decoder to output segmentation predictions.

### B. EM Image Semantic Segmentation

Some studies combine natural and biomedical architectures and extend them to EM image segmentation [20], [21], [22], [23], [25], [32]. [32] used a residual module and pyramid module to fuse multi-scale contextual information for neural cell body and cell nucleus segmentation of EM images. Reference [33] used 3D UNet to predict the affinities of long-range voxel pairs and the affinities of nearest-neighbor voxels and achieved superhuman accuracy on the SNEMI3D Connectomics Challenge. FusionNet [22] and DenseUNet [23] combine UNet [7] with ResNet [34] and DenseNet [35], respectively, for neuronal membrane segmentation. These methods achieve promising results but still lack universality and fail to exploit the intrinsic characteristics of EM images, which limits their practical applicability and further development. In contrast, our architecture serves image characteristics and achieves a universal model for the cellular and subcellular structures of EM images.

### C. Wavelet-Based DNNs for Semantic Segmentation

Based on its powerful frequency and spatial representation capabilities, the wavelet transform has been incorporated into DNNs, and several methods have been explored for semantic segmentation [36], [37], [38], [39], [40]. The common strategies include using the wavelet transform for pre- or post-processing [36], [37] and replacing some layers of CNNs (such as up- and down-sampling layers) with the wavelet transform [38]. However, most of these methods are only suitable for specific segmentation objects, which limits their generalizability and applicability. Reference [41] proposed a symmetric CNN enhanced by the wavelet transform (ALNet) for lane-marking semantic segmentation in aerial imagery. CWNN [42] uses wavelet constrained pooling layers to replace conventional pooling for synthetic aperture radar image segmentation. WaveSNet [43] uses the wavelet transform to extract image details during downs-ampling and uses the inverse transform to recover details during up-sampling. In contrast, we use the wavelet transform to analyze the characteristics of EM images and generate HF prior inputs. We demonstrate that our model outperforms other wavelet-based models in Section IV-D.

### D. Prior Knowledge-Based DNNs for Semantic Segmentation

Our method uses the wavelet transform to generate HF images as extra input, which can be considered to introduce prior knowledge of HF information into the segmentation model. Many studies have proven that introducing prior knowledge into DNNs is an effective way to improve performance [44], [45], [46], [47], [48]. Reference [44] proposed a star-shaped prior loss for skin lesion segmentation, which guarantees a global structure in segmentation predictions. Reference [46] used a discrete Morse theory-based topology as an extra input to segment neural structures. For mitochondria and endolysosome segmentation, since there are many more mitochondria than endolysosomes and they are quite similar in texture, [48] used the available information on mitochondria to improve the segmentation of endolysosomes. Compared with utilizing prior information of segmentation objects (such as topological, anatomical structure, quantity and texture information), our method focuses more on the intrinsic characteristics of images rather than the segmentation objects, mining information from the image itself without being limited to a specific segmentation object type. nnUNet [10] uses dataset characteristics (dataset fingerprints) to adjust the model configuration and training strategies, whereas our method fully integrates EM image characteristics into the architecture design.

## III. Method

We analyze the characteristics of EM images via the wavelet transform in Section III-A. We propose an HF fusion network, GobleNet, to fully exploit these characteristics in Section III-B.

### A. Quantitative Analysis of EM Image Characteristics

Images are essentially discrete non-stationary signals that contain different frequency ranges and spatial location information. The wavelet transform can effectively preserve this information while decomposing images. Specifically, we use the wavelet transform to decompose the raw image $I$ into LF, horizontal HF, vertical HF and diagonal HF components ($LL(I)$, $HL(I)$, $LH(I)$ and $HH(I)$, respectively). They contain LF and different HF information of the raw image. The wavelet transform results of various images are shown in Figure 1 (b), including natural, medical, microscopic and EM images.

We use the sum of the HF components in different directions to generate an HF image $H$ and analyze the differences in $H$ between EM images and other images. $H$ is defined as follows:

$$H = HL(I) + LH(I) + HH(I), \tag{1}$$

and the HF images $H$ corresponding to various images are shown in Figure 1 (c). We find that, compared with other images, the HF images $H$ of EM images contain richer texture and contour details. We use standard deviation convolution to characterize this phenomenon quantitatively. Standard deviation convolution $f(x, y) \star s(x, y)$ uses a convolution kernel $s(x, y)$ with a size of $k$ to convolve the image $f(x, y)$ and calculates the standard deviation of the pixel intensity in the $k \times k$ area. The standard deviation convolution operation is defined as follows:

$$\begin{aligned}\sigma(x, y) &= f(x, y) \star s(x, y) \\ &= \left(\frac{1}{k^2}\sum_{h}^{k}\sum_{w}^{k}(f(x, y) - m^k_{x,y})^2\right)^{\frac{1}{2}},\end{aligned} \tag{2}$$

where $\sigma(x, y)$ represents the convolution result at $(x, y)$, $m^k_{x,y}$ represents the mean pixel intensity of the $k \times k$ convolution area with $(x, y)$ as the center point, $k$ represents the kernel size, and we set $k = 5$ for all analyses.

We can use this convolution to quantitatively calculate the standard deviation of the pixel intensity in each local area of the image. A larger standard deviation indicates that the pixel intensity distribution is more spread out and that the pixel value fluctuation is larger in the local area, i.e., the information richness of the area is greater. Figure 1 (d) shows the information richness heatmap $\sigma_{heatmap}$ of the HF image $H$ for various images. We can see that the pixel intensity fluctuations of $H$ for natural and medical images are small and cannot reflect details such as the texture and boundary contours of segmentation objects, whereas the HF images $H$ of EM images have higher information richness and are concentrated on segmentation boundaries and texture details. Note that the HF images $H$ of microscopic images can also highlight HF details to a certain extent. This may be because the imaging scale, imaging target and imaging method of microscopic images are most similar to those of EM images, so they also reflect some characteristics of EM images.

HF images have rich texture and contour information but also contain large amounts of noise. Higher information richness may be the result of noise interference. We use Gaussian distribution correlation to quantitatively measure the noise intensity distribution. Specifically, for the $k \times k$ convolution area with $(x, y)$ as the center point, the mean $m^k_{x,y}$ and standard deviation $\sigma^k_{x,y}$ of the pixel intensity in the area are first calculated, and then a Gaussian distribution with mean $m^k_{x,y}$ and standard deviation $\sigma^k_{x,y}$ as the convolution kernel is constructed. We normalize the convolved patch and the corresponding convolution kernel to eliminate the interference of uneven pixel intensity. Then, we perform a convolution operation on the area. Consistent with standard deviation convolution, we set $k = 5$ for all analyses. This process is defined as follows:

$$n(x, y) = f(x, y) * g(x, y, m^k_{x,y}, \sigma^k_{x,y}), \tag{3}$$

where $n(x, y)$ represents the Gaussian noise correlation at $(x, y)$. A large $n(x, y)$ represents a large amount of noise. The noise intensity heatmaps $n_{heatmap}$ of the HF images for various images are shown in Figure 1 (e). Compared with other images, the HF images $H$ of EM images are subject to greater noise interference, and the noise is concentrated in non-boundary areas.

HF images can be viewed as consisting of HF details and noise [29], [56], [57]. Furthermore, related studies have shown that noise in biomedical images is generally additive [58], [59]. Therefore, we use subtraction to eliminate noise interference from the information richness heatmap $\sigma_{heatmap}$ to acquire a detailed distribution heatmap $d_{heatmap}$. $d_{heatmap}$ is defined as follows:

$$d_{heatmap} = \sigma_{heatmap} - n_{heatmap}. \tag{4}$$

Figure 1 (f) and (g) show the detailed distribution heatmaps $d_{heatmap}$ of the HF images for various images. We can see

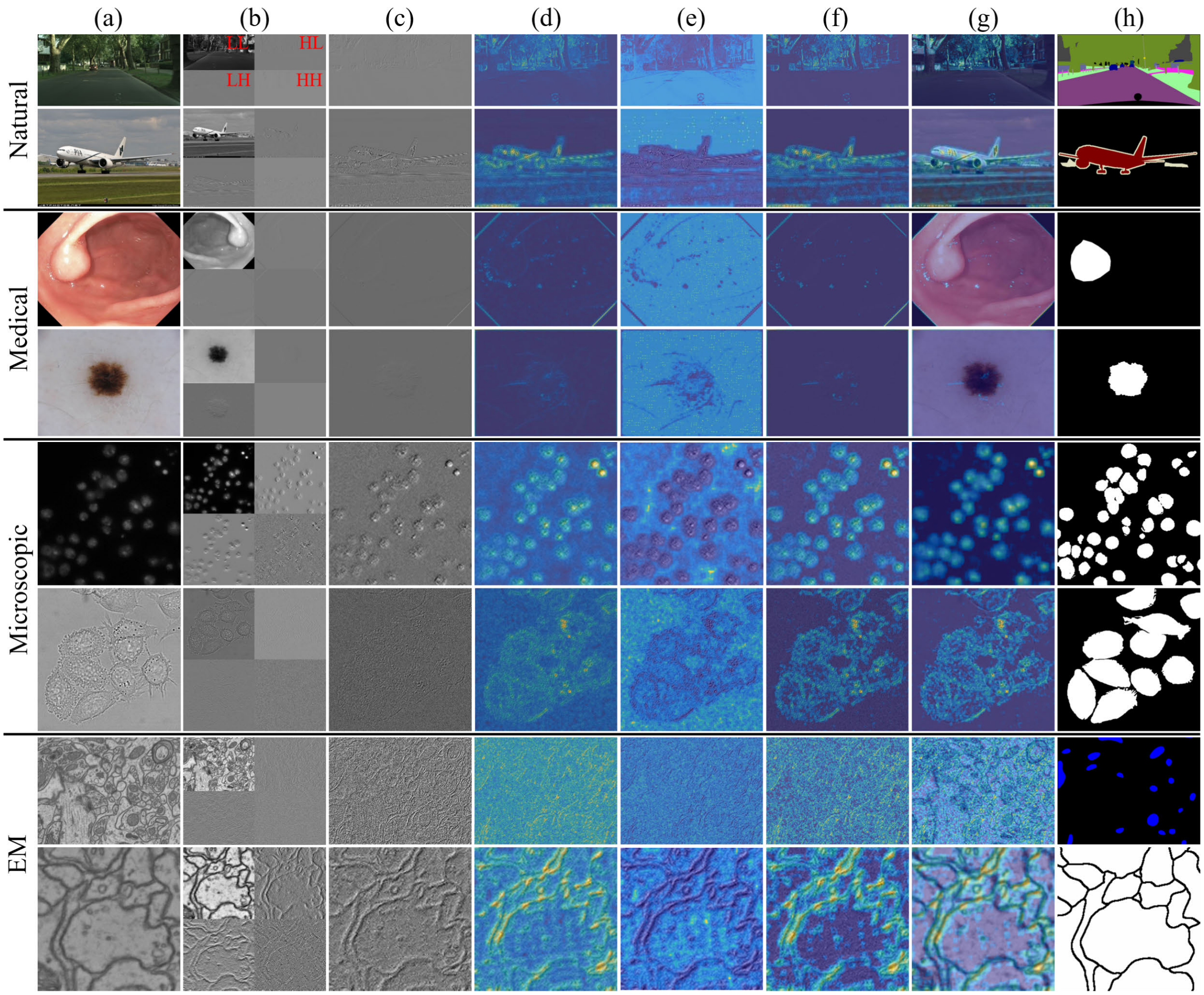


Fig. 1. Qualitative comparison of HF characteristics among natural, medical, microscopic and EM images. From top to bottom: Cityscapes [49], PASCAL VOC 2012 [50], Kvasir-SEG [51], ISIC-2017 [52], 2018 data science Bowl [53], DIC-C2DH-HeLa [54], EPFL [55] and CREMI. We randomly select images from each dataset for analysis and use haar as the wavelet basis. We convert color images into grayscale images and then apply the wavelet transform. (a) Raw images. (b) Wavelet transform results. (c) HF images. (d) Information richness heatmaps. (e) Noise intensity heatmaps. (f) Detailed distribution heatmaps. (g) Detailed distribution heatmaps (overlaid on raw images). (h) Ground truth.

that after noise interference is eliminated, HF images can more prominently emphasize detailed HF information.

To quantitatively evaluate the HF characteristics of different datasets, we normalize $\sigma_{heatmap}$, $n_{heatmap}$, and $d_{heatmap}$ of each image in the dataset and calculate their means $\bar{\sigma}$, $\bar{n}$, and $\bar{d}$, respectively. Then, we calculate the means and standard deviations of $\bar{\sigma}$, $\bar{n}$, and $\bar{d}$ for all the images to characterize the HF information richness (HFIR), noise intensity (NI), and detail richness (DR), respectively, of each dataset quantitatively. Table I compares the HFIR, NI, and DR of datasets with different application scenarios, including natural, medical, microscopic and EM datasets. We find that these metrics are the highest for EM datasets. Furthermore, microscopic datasets are higher than are natural and medical datasets. These conclusions are consistent with those of the previous analysis.

Based on the above analysis, we identify the following characteristic:

TABLE I
QUANTITATIVE COMPARISON OF HFIR, NI AND DR OF DATASETS WITH DIFFERENT APPLICATION SCENARIOS

| Scenario | Dataset | HFIR (%) | NI (%) | DR (%) |
|---|---|---|---|---|
| Natural | Cityscapes | 7.05±1.71 | 3.21±0.54 | 3.84±1.61 |
| | PASCAL VOC 2012 | 16.30±7.09 | 1.92±0.84 | 14.38±7.27 |
| Medical | Kvasir-SEG | 5.13±1.85 | 2.27±0.68 | 2.86±1.40 |
| | ISIC-2017 | 6.97±3.78 | 4.62±1.86 | 2.34±2.86 |
| Microscopic | 2018 Data Science Bowl | 13.59±7.67 | 5.47±2.71 | 8.12±6.72 |
| | DIC-C2DH-HeLa | 18.21±3.21 | 11.32±1.34 | 6.89±2.17 |
| EM | EPFL | 40.47±3.28 | 15.50±1.92 | 24.97±4.41 |
| | CREMI | 36.79±4.48 | 16.09±2.40 | 20.70±5.13 |
| | SNEMI3D | 33.26±4.27 | 10.90±4.17 | 22.35±7.15 |

*1) Characteristic 1:* Compared with other images, the HF components of EM images based on the wavelet transform have richer texture details and clearer object contours but also have more noise.

Noise is often an HF signal, and LF components are relatively less affected by noise [56], [57], [61], [62]. We add

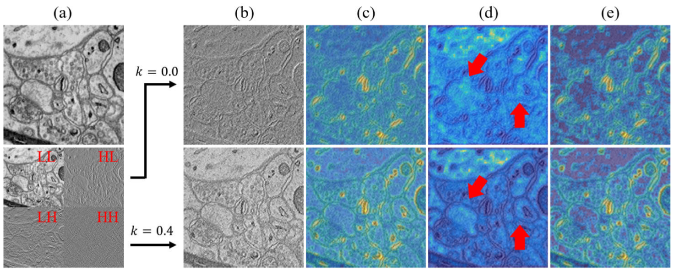


Fig. 2. Comparison of HF images with ($\lambda = 0.4$) and without ($\lambda = 0.0$) LF components on SNEMI3D [60]. (a) Raw image and wavelet transform results. (b) HF images. (c) Information richness heatmaps. (d) Noise intensity heatmaps. (e) Detailed distribution heatmaps. The red arrows highlight the difference in noise intensity.

TABLE II
QUANTITATIVE COMPARISON OF HFIR, NI AND DR OF EM DATASETS WITH DIFFERENT LF WEIGHT $\lambda$ VALUES

| Dataset | $\lambda$ | HFIR (%) | NI (%) | DR (%) |
|---|---|---|---|---|
| EPFL | 0.0 | 40.47±3.28 | 15.50±1.92 | 24.97±4.41 |
| | 0.2 | 39.75±3.18 | 14.03±1.74 | 25.71±4.16 |
| | 0.4 | 38.47±2.81 | 12.25±1.54 | 26.23±3.67 |
| CREMI | 0.0 | 36.79±4.48 | 16.09±2.40 | 20.70±5.13 |
| | 0.2 | 37.13±4.23 | 13.94±2.30 | 23.18±5.03 |
| | 0.4 | 36.88±3.96 | 11.86±2.15 | 25.02±4.89 |
| SNEMI3D | 0.0 | 33.26±4.27 | 10.90±4.17 | 22.35±7.15 |
| | 0.2 | 33.62±4.12 | 9.78±3.62 | 23.83±6.60 |
| | 0.4 | 33.79±3.95 | 8.71±3.13 | 25.08±6.02 |

the LF component $LL(I)$ to the HF image $H$ at the ratio $\lambda$ ($\lambda \in [0, 1]$) to alleviate the interference of noise. The more general form of $H$ is defined as

$$H = HL(I) + LH(I) + HH(I) + \lambda LL(I), \quad (5)$$

Note that $\lambda = 0.0$ indicates that no LF components are added.

Taking SNEMI3D [60] as an example, we compare $\sigma_{heatmap}$, $n_{heatmap}$, and $d_{heatmap}$ for $\lambda = 0.0$ and $\lambda = 0.4$ in Figure 2. Compared with $\lambda = 0.0$ (without $LL(I)$), although some HF details are sacrificed, the noise interference with $\lambda = 0.4$ is significantly reduced. We also compare HFIR, NI, and DR of the EM datasets with different $\lambda$ values in Table II. We find that as $\lambda$ increases, NI decreases, whereas DR increases, which indicates that LF components can effectively alleviate noise interference. We identify the following characteristic based on this analysis:

*2) Characteristic 2:* For EM images, appropriately adding LF components to HF images can alleviate noise interference while maintaining sufficient HF details.

## B. Wavelet-Based HF Fusion Network

*1) Discussion of the Encoder-Decoder Architecture:* The lightweight encoder–decoder design alleviates overfitting caused by the lack of training images. By gradually decreasing and increasing feature map sizes, details can be better restored during encoding and decoding. We optimize this design to retain its advantages and make it more suitable for EM images.

*2) GobletNet:* Guided by the characteristics of EM images, we propose an HF fusion network, GobletNet, which is based on the encoder–decoder architecture. For **Characteristic 1**, we use the HF image $H$ as an extra input and use an extra encoder to extract the rich HF information in $H$. For **Characteristic 2**, we add LF components $LL(I)$ to $H$ at a ratio of $\lambda$ (as shown in Equation (5)) to reduce the negative impact of excessive noise on model training. We demonstrate the performance improvements brought by **Characteristic 1** and **Characteristic 2** and compare different values of the LF weight $\lambda$ and different input strategies for the HF image $H$ in ablation studies in Section IV-F.

Figure 3 shows an overview of the proposed GobletNet. GobletNet consists of three modules: a semantic encoder, an HF detail encoder and a fusion decoder. The semantic encoder uses raw images as inputs to extract the basic semantic information of segmentation objects. The HF detail encoder uses HF images generated by the wavelet transform as inputs to extract HF detail information. The fusion decoder uses semantic features and detail features as inputs to generate segmentation predictions. We use two consecutive residual blocks [34] to form each layer of the encoder and decoder. Each residual block consists of two $3 \times 3$ convolutional layers and a residual connection.

*3) Fusion-Attention Module:* Accurate segmentation requires both semantic information and detail information. The full fusion of semantics and details can produce better results. To achieve their fusion, we introduce a fusion-attention module (FAM) between each corresponding layer of the semantic encoder and the HF detail encoder (as shown in Figure 3). The structure of the FAM is shown in Figure 4. We perform channel concatenation on semantic features and detail features and use $1 \times 1$ convolution to generate fusion features. Then, we use $1 \times 1$ convolution on the fusion features to acquire two different attention features. Both the fusion features and the attention features contain semantic and detail information, but the attention features pay different amounts of attention to them. Finally, we use two attention features to perform element-wise multiplication with the semantic features and detail features to achieve fusion and attention, respectively. We demonstrate the effectiveness of the FAM and compare the performance when FAM is added between different layers in the ablation studies in Section IV-F.

*4) No Complex Design But Effective:* The use of complex models and large-scale data has become the trend in semantic segmentation. However, for biomedicine, the development of segmentation models conflicts with actual needs, whereas the training and fitting of large-scale data conflict with the scarcity of annotated images. The increasingly complex architecture and complicated training process conflict with the simple and efficient segmentation methods required by biomedicine; thus, they are not practical for biomedical research. The increasingly large datasets and higher computing costs conflict with the technical conditions and human resource limitations of ordinary biomedical researchers, making it difficult to support the realization of large-scale training. However, we find an appropriate compromise between these two contradictions. We use the intrinsic characteristics of EM images to drive the architecture design so that the model is simple, effective, and practical while also achieving superior performance at a relatively small training scale.

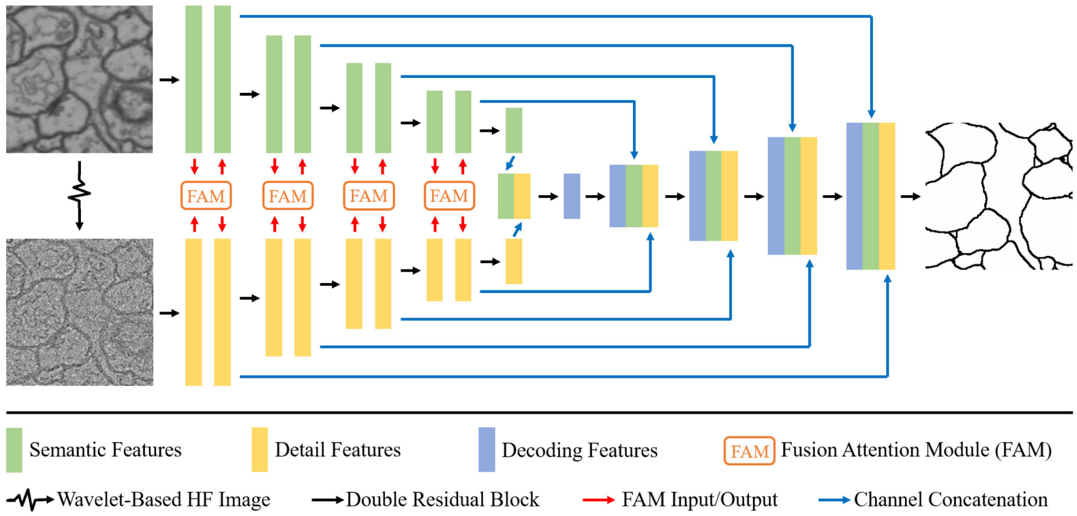


Fig. 3. Overview of GobletNet, which is composed of a semantic encoder, HF detail encoder, FAM and fusion decoder.

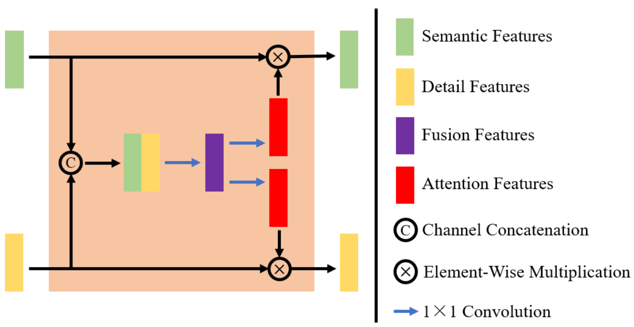


Fig. 4. Architecture of the FAM. The FAM is used for feature fusion between the semantic encoder and the HF detail encoder.

## IV. Experiments

### A. Datasets

We evaluate our model on seven public EM datasets (EPFL [55], CREMI, SNEMI3D [60], UroCell [48], MitoEM [63], Nanowire [64] and BetaSeg [65]).

*1) EPFL:* This is a mitochondrial dataset; it consists of two image stacks, and each stack consists of 165 slices of size $1024 \times 768$. The mitochondria in all the slices are manually annotated by experts. We use the first stack for training and the second stack for testing.

*2) CREMI:* This is a neuronal membrane dataset; it consists of three image stacks for three different types of neurons. Each stack consists of 125 slices of size $1250 \times 1250$. We use the first two stacks for training and the third stack for testing.

*3) SNEMI3D:* This is a neurite dataset; it consists of two image stacks, and each stack consists of 100 slices of size $1024 \times 1024$. We use the first stack for training and the second stack for testing.

*4) UroCell:* This is a multi-organelle dataset of mitochondria, endolysosomes, Golgi apparatuses and fusiform vesicles; it consists of eight sub-volumes of size $256 \times 256 \times 256$. Because the annotations of the Golgi apparatus are approximate without precise boundaries and the annotations of various organelles are distributed in different sub-volumes, we choose four sub-volumes with annotations for mitochondria, endolysosomes, and fusiform vesicles to form our experimental dataset. We use three sub-volumes for training and one for testing.

*5) MitoEM:* This is a mitochondrial dataset; it consists of two volumes from human and rat cortices. Each volume consists of 500 slices of size $4096 \times 4096$. Following [66], we use the first 400 slices of each volume for training and the last 100 slices for testing.

*6) Nanowire:* This is a peptide/protein nanowire dataset; it consists of nanowire images with four morphologies: bundle, singular, dispersed and network. Each morphology corresponds to 100 images of size $1024 \times 1024$. Expert annotations are provided for the dispersed and network morphologies, and we use them to form our experimental dataset. We randomly select 100 images for training and use the remaining 100 images for testing.

*7) BetaSeg:* This is a multi-organelle dataset; it images seven islet $\beta$ cells, and generates corresponding image stacks. Expert annotations are provided for the mitochondria, Golgi apparatus, cell nuclei and insulin secretory granules. We use five stacks for training and two for testing.

For CREMI, SNEMI3D, MitoEM, Nanowire and BetaSeg, because large image sizes are not convenient for training, we use a sliding window to crop the raw images to form a new experimental dataset. Additional details on the imaging conditions and experimental setup for these datasets are presented in Table III.

*8) Why Choose These Datasets?:* According to Table III, these datasets cover different imaging conditions, segmentation objects, image sizes and data scales. Evaluating model performance is more representative and convincing.

### B. Evaluation Metrics

We use the Jaccard index (Jaccard), Dice coefficient (Dice), 95th-percentile Hausdorff distance (95HD), and average surface distance (ASD) as performance metrics to evaluate the segmentation results. Jaccard and Dice emphasize pixel-wise

TABLE III

IMAGING CONDITIONS AND EXPERIMENTAL SETUPS OF SEVEN DATASETS. - INDICATES THAT THE VOXEL SIZE IS NOT MENTIONED IN THE ORIGINAL PAPER. M-MITOCHONDRIA, NM-NEURONAL MEMBRANE, E-ENDOLYSOSOME, F-FUSIFORM VESICLE, N-NANOWIRE, G-GOLGI APPARATUS, CN-CELL NUCLEUS, AND ISG-INSULIN SECRETORY GRANULE

| Dataset | Imaging Conditions | | | | Experimental Setup | | | |
|---|---|---|---|---|---|---|---|---|
| | Acquisition | Region | Voxel Size (nm) | Volume Size | Annotation | Preprocessed Size | # Training | # Test |
| EPFL | FIB-SEM | Mouse - Hippocampus | 5×5×5 | 1024×768×165 | M | 1024×768 | 165 | 165 |
| CREMI | ssTEM | Drosophila & Adult Fly - Brain | 4×4×40 | 1250×1250×125 | NM | 256×256 | 3575 | 3075 |
| SNEMI3D | ssSEM | Mouse - Neocortex | 3×3×30 | 1024×1024×100 | NM | 256×256 | 839 | 1600 |
| UroCell | FIB-SEM | Mouse - Urothelial Cell | 16×16×15 | 256×256×256 | M, E, F | 256×256 | 768 | 256 |
| MitoEM | ssSEM | Rat & Human - Cortex | 8×8×30 | 4096×4096×500 | M | 1024×1024 | 3964 | 3200 |
| Nanowire | TEM | Escherichia Coli - Nanowire | - | 2048×2048 | PN | 512×512 | 234 | 400 |
| BetaSeg | FIB-SEM | Mice - Islet | 16×16×16 | 725×545×853∼1082×758×1097 | M, G, CN, ISG | 256×256 | 3740 | 2735 |

TABLE IV

COMPARISON WITH STATE-OF-THE-ART MODELS ON THE EPFL, CREMI, SNEMI3D AND UROCELL TEST SETS. * INDICATES A LIGHTWEIGHT MODEL. ‡ INDICATES TRAINING FOR 1000 EPOCHS. - INDICATES THAT TRAINING FAILED. BOLD INDICATES THE BEST PERFORMANCE IN EACH SCENARIO. RED INDICATES THE BEST PERFORMANCE AMONG ALL MODELS

| Scenario | Model | EPFL | | | | CREMI | | | | SNEMI3D | | | | UroCell | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ |
| Natural | Deeplab V3+ [2] | 72.23 | 83.88 | 19.95 | 3.50 | 56.29 | 72.03 | 4.68 | 1.17 | 57.77 | 73.23 | 5.09 | 1.44 | 59.77 | 71.59 | 21.24 | 6.95 |
| | Res-UNet [67] | **78.70** | **88.08** | **12.93** | **2.31** | 71.39 | 83.31 | 4.66 | 0.94 | 61.09 | 75.85 | 4.61 | 1.26 | **71.94** | **82.82** | **17.55** | 7.26 |
| | $U^2$-Net [68] | - | - | - | - | **75.63** | **86.13** | **3.93** | **0.74** | 60.05 | 75.04 | 4.23 | 1.26 | 67.42 | 78.96 | 21.97 | **5.09** |
| | $U^2$-Net* [68] | 78.10 | 87.71 | 19.32 | 2.67 | 71.21 | 83.18 | 4.82 | 0.88 | **61.37** | **76.06** | **4.15** | **1.21** | 67.71 | 79.01 | 22.29 | 5.21 |
| Medical | UNet [7] | 73.75 | 84.89 | 22.19 | 3.42 | 72.43 | 84.01 | 4.34 | 0.82 | 61.06 | 75.82 | 4.23 | **1.20** | 68.79 | 79.84 | **16.24** | 6.29 |
| | UNet++ [8] | 77.24 | 87.16 | **17.26** | **2.71** | 71.24 | 83.21 | 3.72 | 0.74 | 60.63 | 75.49 | 6.19 | 1.59 | **71.20** | **82.01** | 17.78 | 5.68 |
| | Att-UNet [69] | 69.44 | 81.96 | 29.11 | 4.11 | 71.83 | 83.61 | 4.15 | 0.81 | 61.23 | 76.02 | 4.54 | 1.30 | 65.65 | 76.64 | 16.41 | 5.60 |
| | UNet 3+ [9] | 71.53 | 83.40 | 28.36 | 3.97 | 70.46 | 82.67 | 6.40 | 1.26 | 60.57 | 75.44 | 4.91 | 1.34 | 66.46 | 77.94 | 19.21 | 7.25 |
| | SwinUNet [70] | - | - | - | - | 62.38 | 76.83 | 9.31 | 1.75 | 56.32 | 72.05 | 7.25 | 2.02 | 55.03 | 66.95 | 39.91 | 10.27 |
| | nnUNet [10] | 77.12 | 87.08 | 24.15 | 3.15 | 70.19 | 82.48 | 5.08 | 1.09 | 57.48 | 73.00 | 9.79 | 2.69 | 69.77 | 80.64 | 17.92 | **5.00** |
| | nnUNet‡ [10] | 78.76 | 88.12 | 21.01 | 2.83 | 74.96 | 85.6 | **3.64** | 0.66 | 57.64 | 73.13 | 9.89 | 2.66 | 68.62 | 79.43 | 22.19 | 6.82 |
| | XNet [29] | **80.02** | **88.90** | **12.84** | **2.31** | **79.23** | **88.41** | 3.66 | **0.61** | **63.50** | **77.68** | **4.16** | 1.22 | - | - | - | - |
| Wavelet | ALNet [41] | **72.56** | **84.10** | **22.73** | **3.68** | 67.88 | 80.87 | 5.25 | 1.09 | 60.11 | 75.09 | 7.18 | 1.99 | 67.30 | 78.75 | 17.27 | **4.94** |
| | MWCNN [40] | 67.02 | 80.25 | 29.96 | 4.40 | 69.02 | 81.67 | **5.06** | **0.99** | 57.86 | 73.31 | 6.33 | 1.63 | **71.97** | **82.84** | 16.77 | 5.28 |
| | WDS [39] | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - | - |
| | WaveSNet [43] | 71.00 | 83.04 | 24.95 | 3.69 | **69.80** | **82.21** | 6.08 | 1.19 | **61.27** | **75.98** | **3.52** | **1.08** | 70.53 | 81.53 | **15.41** | 5.00 |
| General | SAM [71] | **68.23** | **81.11** | **42.65** | **5.90** | **61.96** | **76.51** | **10.56** | **1.85** | **47.05** | **63.99** | **19.12** | **4.39** | **58.25** | **65.71** | **27.89** | **7.86** |
| EM | DCR [21] | 74.15 | 85.16 | 19.30 | 3.09 | 69.45 | 81.97 | 6.85 | 1.31 | 59.70 | 74.77 | 5.29 | 1.40 | 67.75 | 79.36 | 20.86 | 5.66 |
| | FusionNet [22] | 76.55 | 86.71 | 19.77 | 2.93 | 69.59 | 82.07 | 5.36 | 1.06 | 61.64 | 76.27 | 5.20 | 1.40 | 65.81 | 77.23 | 15.83 | 4.82 |
| | GobletNet (Ours) | **81.66** | **89.91** | **12.04** | **2.12** | **80.17** | **89.00** | **3.02** | **0.54** | **64.18** | **78.18** | **3.58** | **1.04** | **74.42** | **84.61** | **13.47** | **4.45** |

accuracy, whereas 95HD and ASD emphasize boundary accuracy. These metrics are widely used for benchmarking performance in biomedical image segmentation.

## C. Implementation Details

We implement our model via PyTorch. Training and inference of all the models are performed on four NVIDIA GeForce RTX3090. We use SGD with momentum to train the models; the momentum is set to 0.9, and the weight decay is set to 0.00005. The batch size is set to 16. The number of epochs is set to 200. The initial learning rate is set to 0.5, and the learning rate decays by 0.5 every 50 epochs. We use Dice loss [11] as the loss function. Flip, rotation, and transposition are used for data augmentation. Considering GPU memory limitations and training efficiency, for EPFL, MitoEM and Nanowire, the input images are resized to 256 × 256, and for CREMI, SNEMI3D, UroCell and BetaSeg, the input images are resized to 128 × 128.

## D. Comparison With State-of-the-Art Models

We compare GobletNet extensively with state-of-the-art models, including natural, medical, wavelet, general and EM models, in different application scenarios.

Table IV and Table V show the comparison results on seven datasets. GobletNet outperforms previous state-of-the-art models by a large margin; this may be because our model appropriately and fully exploits the characteristics of EM images. Specifically, the wavelet transform highlights HF information in images. Although noise is present, this HF information (such as texture and contours) is beneficial for accurate identification of the boundaries of segmentation objects while alleviating the interference of impurities that are similar to segmentation objects. Our model uses an extra encoder to extract these HF features and continuously fuses them with features from the raw images, ultimately achieving superior pixel-wise accuracy and boundary accuracy. We note that XNet also achieves positive results, which may be due to its consistent learning of LF and HF images. We also find that fine-tuned general models (SAM) perform poorly; this may be because general models are trained on a large amount of data, but these data contain very few EM images.

## E. Qualitative Results

Figure 5 shows various qualitative results of different models, including SAM [71], Deeplab V3+ [2], UNet 3+ [9], FusionNet [22], WaveSNet [43], UNet [7], nnUNet [10], and GobletNet. Based on the efficient utilization and fusion of HF information in EM images, GobletNet achieves better pixel-wise accuracy and contour accuracy while effectively avoiding the interference of non-target noise.

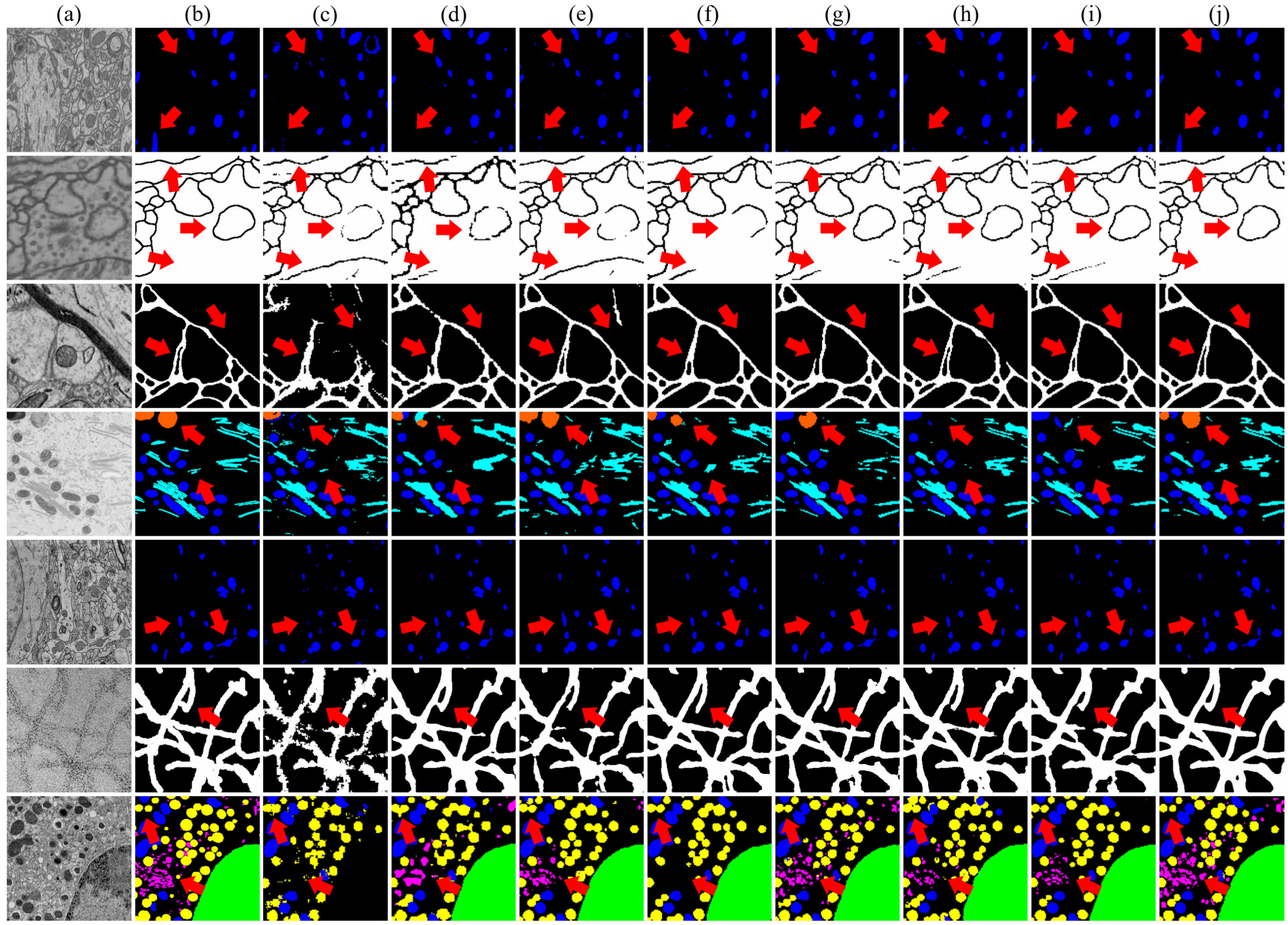


Fig. 5. Qualitative results of different models. From top to bottom: EPFL, CREMI, SNEMI3, UroCell, MitoEM, Nanowire and BetaSeg. (a) Raw images. (b) Ground truth. (c) SAM. (d) Deeplab V3+. (e) UNet 3+. (f) FusionNet. (g) WaveSNet. (h) UNet. (i) nnUNet. (j) GobletNet. The red arrows highlight the differences among the results.

TABLE V
COMPARISON WITH STATE-OF-THE-ART MODELS ON THE MITOEM, NANOWIRE AND BETASEG TEST SETS. * INDICATES A LIGHTWEIGHT MODEL. ‡ INDICATES TRAINING FOR 1000 EPOCHS. - INDICATES THAT TRAINING FAILED. BOLD INDICATES THE BEST PERFORMANCE IN EACH SCENARIO. RED INDICATES THE BEST PERFORMANCE AMONG ALL MODELS

| Scenario | Model | MitoEM | | | | Nanowire | | | | BetaSeg | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ |
| Natural | Deeplab V3+ [2] | 79.12 | 88.34 | 18.27 | 2.86 | 71.51 | 83.39 | 11.05 | 2.91 | 57.93 | 70.43 | 44.76 | 18.01 |
| | Res-UNet [67] | 83.76 | 91.17 | **11.47** | **1.94** | 70.53 | 82.72 | 10.62 | 2.76 | 60.81 | 72.86 | 53.45 | 23.04 |
| | $U^2$-Net [68] | **84.21** | **91.43** | 13.92 | 2.20 | 71.67 | 83.50 | 10.35 | **2.58** | **65.75** | **75.97** | **36.61** | **14.83** |
| | $U^2$-Net* [68] | 83.94 | 91.27 | 12.09 | 1.99 | **71.75** | **83.55** | **10.23** | 2.64 | 62.64 | 72.20 | 39.43 | 16.32 |
| Medical | UNet [7] | 83.98 | 91.29 | 11.38 | 1.95 | 71.39 | 83.31 | 10.18 | 2.64 | 64.00 | 74.07 | 42.14 | 17.64 |
| | UNet++ [8] | 84.07 | 91.35 | 11.05 | 1.93 | 71.89 | 83.64 | **9.68** | **2.54** | 61.13 | 73.39 | 43.37 | 17.67 |
| | Att-UNet [69] | 83.31 | 90.90 | 10.92 | 1.92 | 71.33 | 83.26 | 11.55 | 2.83 | 66.47 | 76.65 | 40.24 | 16.78 |
| | UNet 3+ [9] | 83.76 | 91.17 | 11.47 | 1.94 | 70.53 | 82.72 | 10.62 | 2.76 | 60.81 | 72.86 | 53.45 | 23.04 |
| | SwinUNet [70] | 71.12 | 83.13 | 22.93 | 3.77 | 67.49 | 80.59 | 12.35 | 3.04 | 39.16 | 47.37 | 70.00 | 30.26 |
| | nnUNet [10] | 83.28 | 90.88 | 13.04 | 2.14 | **72.45** | **84.02** | 10.21 | 2.61 | 68.11 | 78.46 | **30.20** | **11.57** |
| | nnUNet‡ [10] | 83.35 | 90.92 | 13.11 | 2.14 | 72.43 | 84.01 | 10.45 | 2.60 | **68.37** | **78.08** | 31.09 | 12.22 |
| | XNet [29] | **87.61** | **93.40** | **10.43** | **1.69** | 72.37 | 83.97 | 10.01 | 2.57 | 64.36 | 71.03 | 57.32 | 26.51 |
| Wavelet | ALNet [41] | 82.42 | 90.36 | 13.10 | 2.17 | 70.33 | 82.58 | 12.35 | 2.91 | 58.86 | 67.22 | 61.13 | 27.79 |
| | MWCNN [40] | 82.82 | 90.60 | 14.30 | 2.32 | **71.57** | **83.43** | 10.37 | 2.68 | - | - | - | - |
| | WDS [39] | - | - | - | - | - | - | - | - | - | - | - | - |
| | WaveSNet [43] | **83.39** | **90.95** | **11.91** | **2.03** | 71.37 | 83.29 | **9.76** | **2.63** | **66.28** | **76.88** | **35.62** | **14.00** |
| General | SAM [71] | **75.19** | **85.84** | **25.38** | **3.95** | **61.22** | **75.95** | **21.76** | **4.38** | **41.57** | **48.99** | **97.76** | **46.82** |
| EM | DCR [21] | 79.68 | 88.69 | 16.33 | 2.60 | 71.73 | 83.54 | **9.32** | **2.56** | - | - | - | - |
| | FusionNet [22] | **83.26** | **90.86** | **11.94** | **2.04** | **72.11** | **83.79** | 10.01 | 2.60 | **57.66** | **66.16** | **58.28** | **26.47** |
| | GobletNet (Ours) | **88.31** | **93.79** | **9.62** | **1.61** | **72.50** | **84.06** | 9.60 | 2.60 | **69.74** | **80.25** | **28.09** | **11.33** |

## F. Ablation Studies

To verify the effectiveness of each component, we perform the following ablation studies on EPFL, CREMI, SNEMI3D and UroCell.

### 1) Comparison of Different Values of the LF Weight $\lambda$:

We compare the performance under different values of the LF weight $\lambda$ in Table VI. We find that the performance differences among different $\lambda$ values are small, and in general,

TABLE VI
COMPARISON OF DIFFERENT VALUES OF THE LF WEIGHT $\lambda$. THE WAVELET BASIS IS DB2

| $\lambda$ | EPFL | | | | CREMI | | | | SNEMI3D | | | | UroCell | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ |
| 0.0 | 79.48 | 88.57 | 15.05 | 2.50 | 79.94 | 88.85 | **3.08** | **0.53** | 63.81 | 77.91 | **3.69** | **1.09** | **71.83** | **82.69** | 18.70 | 5.43 |
| 0.1 | 80.02 | 88.90 | 12.84 | **2.31** | 79.81 | 88.77 | 3.69 | 0.62 | 63.54 | 77.70 | 4.77 | 1.23 | 68.62 | 79.43 | 22.19 | 6.82 |
| 0.2 | 79.32 | 88.47 | 15.92 | 2.68 | 79.88 | 88.82 | 3.14 | 0.54 | **63.88** | **77.96** | 4.80 | 1.30 | 70.06 | 80.83 | 17.49 | 6.56 |
| 0.3 | **81.18** | **89.61** | 14.83 | 2.44 | **80.10** | **88.95** | 3.60 | 0.63 | 63.85 | 77.94 | 4.86 | 1.30 | 67.32 | 78.19 | **15.13** | **4.18** |
| 0.4 | 80.08 | 88.94 | 16.59 | 2.55 | 79.79 | 88.76 | 3.46 | 0.61 | 63.80 | 77.90 | 5.04 | 1.34 | 70.38 | 81.35 | 15.58 | 6.11 |
| 0.5 | 80.56 | 89.23 | **12.51** | 2.33 | 79.97 | 88.87 | 3.46 | 0.61 | 63.85 | 77.94 | 4.16 | 1.15 | 70.51 | 81.45 | 17.76 | 5.53 |

TABLE VII
COMPARISON OF DIFFERENT WAVELET BASES

| Wavelet | EPFL | | | | CREMI | | | | SNEMI3D | | | | UroCell | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ |
| Haar | **81.66** | **89.91** | 12.04 | **2.12** | 79.87 | 88.81 | **2.95** | **0.50** | 64.17 | 78.17 | 4.04 | 1.11 | 69.46 | 79.71 | 16.32 | 5.03 |
| Dmey | 79.50 | 88.58 | **11.25** | 2.22 | 76.12 | 86.44 | 3.77 | 0.66 | 63.54 | 77.71 | 3.69 | 1.08 | 65.53 | 76.75 | 15.63 | 5.94 |
| Coif1 | 79.37 | 88.50 | 13.39 | 2.30 | **80.17** | **89.00** | 3.02 | 0.54 | 63.71 | 77.83 | 3.50 | **1.02** | 67.39 | 78.33 | 15.35 | **4.03** |
| Bior1.5 | 81.12 | 89.58 | 11.30 | 2.13 | 79.90 | 88.83 | 3.33 | 0.56 | 63.92 | 77.99 | **3.36** | 1.11 | **74.42** | **84.61** | **13.47** | 4.45 |
| Bior2.4 | 79.10 | 88.33 | 18.91 | 2.69 | 79.95 | 88.86 | 3.48 | 0.59 | **64.18** | **78.18** | 3.58 | 1.04 | 67.11 | 77.46 | 16.96 | 6.42 |
| Db2 | 81.18 | 89.61 | 14.83 | 2.44 | 80.10 | 88.95 | 3.60 | 0.63 | 63.88 | 77.96 | 4.80 | 1.30 | 71.83 | 82.69 | 18.70 | 5.43 |

TABLE VIII
COMPARISON OF DIFFERENT INPUT STRATEGIES. $L$ AND $H$ INDICATE THE LF COMPONENT AND HF IMAGE, RESPECTIVELY, GENERATED BY THE WAVELET TRANSFORM. $H^2$ INDICATES USING THE SECOND-LEVEL WAVELET TRANSFORM TO GENERATE THE HF IMAGE. W/O INDICATES THAT NO EXTRA INPUT IS USED. CHANNEL INDICATES USING EXTRA INPUT AS AN EXTRA CHANNEL OF THE RAW IMAGES THAT ARE INPUT INTO THE MODEL. BRANCH INDICATES INPUTTING THE IMAGES INTO AN EXTRA ENCODING BRANCH (I.E., AN HF DETAIL ENCODER) AND USING FAM TO FUSE FEATURES

| Input Strategy | | EPFL | | | | CREMI | | | | SNEMI3D | | | | UroCell | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ |
| w/o | | 75.55 | 86.07 | 20.96 | 3.26 | 72.62 | 84.14 | 4.55 | 0.80 | 61.58 | 76.22 | 4.84 | 1.31 | 70.62 | 81.52 | 20.51 | 5.51 |
| $L$ | Channel | 74.51 | 85.40 | 21.22 | 3.33 | 73.39 | 84.65 | 3.89 | 0.75 | 61.31 | 76.01 | 5.71 | 1.51 | 66.62 | 77.20 | 15.68 | 6.58 |
| | Branch | 79.09 | 88.33 | 15.26 | 2.56 | 69.55 | 82.04 | 4.06 | 0.81 | 61.32 | 76.02 | 4.69 | 1.26 | 68.52 | 79.98 | 19.09 | 5.60 |
| $H$ | Channel | 76.05 | 86.40 | 23.25 | 3.47 | 79.31 | 88.46 | 3.67 | 0.63 | 63.39 | 77.59 | 4.77 | 1.26 | 73.15 | 83.54 | 18.71 | 5.13 |
| | Branch | **81.66** | **89.91** | **12.04** | **2.12** | **80.17** | **89.00** | **3.02** | **0.54** | **64.18** | **78.18** | 3.58 | 1.04 | **74.42** | **84.61** | 13.47 | 4.45 |
| $H^2$ | Channel | 76.58 | 86.74 | 22.78 | 3.35 | 79.73 | 88.72 | 3.81 | 0.62 | 62.04 | 76.58 | 3.72 | 1.11 | 68.44 | 79.59 | 17.47 | **3.74** |
| | Branch | 81.63 | 89.89 | 12.63 | 2.21 | 79.89 | 88.82 | 3.58 | 0.61 | 64.14 | 78.15 | **3.23** | **1.00** | 73.32 | 83.66 | **13.44** | 3.88 |

TABLE IX
COMPARISON OF UNET WITH AND WITHOUT HF IMAGES AS EXTRA INPUT. RAW INDICATES ONLY USING RAW IMAGES AS INPUT. RAW+H INDICATES USING HF IMAGES AS AN EXTRA CHANNEL OF THE RAW IMAGES THAT ARE INPUT INTO THE MODEL. B-JACCARD AND B-DICE INDICATE THE JACCARD INDEX AND DICE COEFFICIENT OF THE BOUNDARY PIXELS OF SEGMENTATION OBJECTS

| Model | Dataset | Input | Jaccard ↑ | B-Jaccard ↑ | Dice ↑ | B-Dice ↑ | 95HD ↓ | ASD ↓ |
|---|---|---|---|---|---|---|---|---|
| UNet | EPFL | Raw | 73.75 | 21.03 | 84.89 | 34.76 | **22.19** | 3.42 |
| | | Raw+H | **75.35** | **24.72** | **85.94** | **39.64** | 22.22 | **3.18** |
| | CREMI | Raw | 72.43 | 62.32 | 84.01 | 76.79 | 4.34 | 0.82 |
| | | Raw+H | **79.35** | **73.18** | **88.49** | **84.51** | **3.72** | **0.61** |
| | SNEMI3D | Raw | 61.06 | 20.94 | 75.82 | 34.63 | 6.24 | 1.54 |
| | | Raw+H | **62.57** | **22.46** | **76.98** | **36.69** | **4.23** | **1.20** |
| | UroCell | Raw | 68.79 | 47.26 | 79.84 | 59.40 | 16.24 | 6.29 |
| | | Raw+H | **70.84** | **48.48** | **81.47** | **60.29** | **14.19** | **4.04** |

$\lambda = 0.0 \sim 0.3$ results in better performance. Based on the analysis in Section III-A, this may be because proper fusion of LF images with HF images can reduce noise and is beneficial for model training. Based on the above experiments, we select $\lambda = 0.3$, 0.3, 0.2 and 0.0 for the four datasets and apply these values in all related experiments.

*2) Comparison of Wavelet Bases:* Table VII compares the performance of different wavelet bases, including Haar, Dmey, Coif1, Bior1.5, Bior2.4 and Db2, on four datasets. We find that EPFL, CREMI and SNEMI3D are relatively robust to the wavelet basis and that the performance differences among different wavelet bases are small. In contrast, UroCell is more sensitive to the wavelet basis, and the performance difference between the best (Bior1.5) and the worst (Demy) cases is 8.89% for Jaccard, 7.86% for Dice, 5.23 pixels for 95HD and 2.39 pixels for ASD. This may be because UroCell is a multi-organelle dataset, and the various organelle contours between different wavelet-based HF images are quite different. We also find that for EPFL and SNEMI3D, Haar and Bior2.4 achieve better results, respectively. For CREMI, Coif1 has better pixel-wise accuracy, and Haar has better boundary accuracy. Based on the above experiments, we select Haar, Coif1, Bior2.4 and Bior1.5 as the wavelet bases for the four datasets and apply them to all related experiments.

*3) Effectiveness of HF Images:* To demonstrate the effectiveness of wavelet-based HF images, we compare different input strategies in Table VIII. Compared with the use of no extra input, the use of only the HF image as an extra channel of the raw images can significantly improve performance. Furthermore, we find that using an extra encoding branch (i.e., an HF detail encoder) to extract features can further improve performance. We also use HF images for UNet, and the results are shown in Table IX. As in the experiments in Table VIII, only using HF images as an extra channel can improve performance by a large margin. These experiments strongly demonstrate the effectiveness of HF images. Furthermore, we also compare the LF component

TABLE X

Comparison of Adding the FAM Into Different Layers. w/o Indicates That the FAM Is Not Used in the Model. Shallow, Middle and Deep Indicate Adding the FAM Into the Shallow Layer (First layer), Middle Layer (Second and Third Layer) and Deep Layer (Fourth layer), Respectively, of the Encoder. Full Indicates Adding the FAM Into All Layers of the Encoder

| FAM | EPFL | | | | CREMI | | | | SNEMI3D | | | | UroCell | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ |
| w/o | 78.83 | 88.16 | 15.37 | 2.47 | 79.58 | 88.63 | 3.38 | 0.56 | 63.69 | 77.82 | 4.09 | 1.14 | 72.18 | 82.60 | 18.00 | 6.20 |
| Shallow | 81.51 | 89.82 | 14.32 | 2.38 | 79.75 | 88.74 | 3.41 | 0.56 | 63.95 | 78.01 | 3.78 | 1.07 | 72.51 | 82.85 | 14.14 | **4.10** |
| Middle | 78.48 | 87.94 | 16.59 | 2.66 | 79.38 | 88.51 | 4.20 | 0.69 | 63.80 | 77.90 | 3.85 | 1.07 | 70.39 | 81.01 | 18.24 | 4.47 |
| Deep | 77.31 | 87.20 | 18.86 | 2.92 | 79.40 | 88.52 | 3.92 | 0.67 | 63.55 | 77.71 | **3.13** | **1.00** | 73.74 | 84.00 | 23.85 | 8.12 |
| Full | **81.66** | **89.91** | **12.04** | **2.12** | **80.17** | **89.00** | **3.02** | **0.54** | **64.18** | **78.18** | 3.58 | 1.04 | **74.42** | **84.61** | **13.47** | 4.45 |

TABLE XI

Comparison of Model Size and Computational Cost. $^{+}$ Indicates Expanding the Model Size by Increasing the Number of Feature Channels. $^{-}$ and $^{--}$ Indicate Reducing the Model Size by Using Half and a Quarter of the Feature Channels, Respectively

| Model | Params | MACs | EPFL | | | | CREMI | | | | SNEMI3D | | | | UroCell | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ |
| UNet$^{+}$ | 138M | 65G | 74.09 | 85.12 | 22.48 | 3.38 | 72.87 | 84.31 | 6.57 | 1.24 | 61.21 | 75.94 | 4.25 | 1.23 | 70.62 | 81.59 | 17.00 | 6.00 |
| UNet 3+$^{+}$ | 83M | 69G | 67.43 | 80.55 | 29.67 | 4.46 | 74.20 | 85.19 | 3.34 | 0.67 | 60.32 | 75.25 | 4.28 | 1.20 | 66.02 | 77.15 | 24.02 | 7.35 |
| WaveSNet$^{+}$ | 118M | 41G | 74.38 | 85.31 | 20.82 | 3.19 | 71.79 | 83.58 | 4.53 | 0.83 | 61.23 | 75.95 | 3.73 | 1.11 | 71.50 | 82.55 | 17.00 | 5.98 |
| FusionNet$^{+}$ | 123M | 47G | 75.48 | 86.03 | 18.28 | 2.93 | 71.18 | 83.17 | 3.37 | 0.71 | 61.93 | 76.49 | 3.59 | 1.05 | 71.75 | 82.39 | 15.63 | 5.35 |
| GobletNet | 133M | 46G | **81.66** | **89.91** | **12.04** | **2.12** | **80.17** | **89.00** | **3.02** | **0.54** | **64.18** | **78.18** | **3.58** | **1.04** | **74.42** | **84.61** | **13.47** | **4.45** |
| UNet | 35M | 16G | 73.75 | 84.89 | 22.19 | 3.42 | 72.43 | 84.01 | 4.34 | 0.82 | 61.06 | 75.82 | 4.23 | 1.20 | 68.79 | 79.84 | 16.24 | 6.29 |
| WaveSNet | 29M | 11G | 71.00 | 83.04 | 24.95 | 3.69 | 69.80 | 82.21 | 6.08 | 1.19 | 61.27 | 75.98 | **3.52** | 1.08 | 70.53 | 81.53 | **15.41** | **5.00** |
| GobletNet$^{-}$ | 33M | 11G | **80.36** | **89.11** | **16.03** | **2.54** | **79.53** | **88.60** | **3.69** | **0.69** | **63.77** | **77.87** | 3.55 | **1.04** | **71.93** | **82.60** | 15.74 | 6.34 |
| UNet 3+ | 2M | 3G | 71.53 | 83.40 | 28.36 | 3.97 | 70.46 | 82.67 | 6.40 | 1.26 | 60.57 | 75.44 | **4.91** | 1.34 | 66.46 | 77.94 | 19.21 | 7.25 |
| FusionNet | 5M | 2G | 76.55 | 86.71 | 19.77 | 2.93 | 69.59 | 82.07 | 5.36 | 1.06 | 61.64 | 76.27 | 5.20 | 1.40 | 65.81 | 77.23 | **15.83** | 4.82 |
| GobletNet$^{--}$ | 8M | 3G | **80.21** | **89.02** | **16.24** | **2.51** | **79.07** | **88.31** | **3.15** | **0.59** | **63.51** | **77.68** | 4.92 | **1.33** | **70.96** | **81.61** | 17.85 | **4.80** |

and the second-level wavelet transform in Table VIII. The introduction of the LF component has a negative impact, which indirectly proves that HF detail information is crucial for EM image segmentation. Using the second-level wavelet transform to generate an HF image as input does not significantly improve performance, which may be because the second-level transform loses the HF components of the first-level transform.

*4) Effectiveness of the FAM:* To demonstrate the effectiveness of the FAM, we compare the performance with the FAM at different layers and without it. The comparison results are shown in Table X. Compared with without the FAM, introducing the FAM to the shallow, middle and deep layers of the encoder can improve performance, whereas introducing the FAM to all the layers of the encoder achieves the best results.

*5) Comparison of Model Size and Computational Cost:* To demonstrate that the performance improvement originates from the well-designed components rather than the additional parameters of GobletNet, we compare the performance of models of similar size and computational cost in Table XI. Specifically, we expand UNet, UNet 3+, WaveSNet and FusionNet to similar numbers of parameters (Params) and multiply accumulate operations (MACs) as GobletNet. In some cases, this has positive effects, but they do not reach the performance of GobletNet. Furthermore, we reduce the number of channels of GobletNet to half and a quarter. Compared with models of similar size and computational cost, these lightweight versions of GobletNet still have superior performance. These experiments strongly demonstrate that the designs of various components are the keys to the performance improvement rather than the number of model parameters.

*6) Effectiveness of Components:* To demonstrate the effectiveness of different components, we conduct step-by-step ablation studies on four datasets, and the results are shown in Table XII. We take SNEMI3D as an example to show the effects of each component. Using only the raw images that are input into the conventional encoder-decoder architecture, we achieve a baseline performance of 61.58% for Jaccard, 76.22% for Dice, 4.84 pixels for 95HD and 1.31 pixels for ASD. Using HF images as an extra channel of raw images improves the baseline by 1.81% in terms of Jaccard, 1.37% in terms of Dice, 0.07 pixels in terms of 95HD and 0.05 pixels in terms of ASD. An extra encoding branch (i.e., an HF detail encoder) is used to further improve the baseline to 63.69% for Jaccard, 77.82% for Dice, 4.09 pixels for 95HD and 1.14 pixels for ASD. By introducing the FAM in the encoder, we finally achieve a performance of 64.18% for Jaccard, 78.18% for Dice, 3.58 pixels for 95HD and 1.04 pixels for ASD.

*7) Qualitative Comparison of the Class Activation Map:* To further understand why our model outperforms other models, we compare the class activation map (CAM) [72] of the encoding and decoding layers between UNet and GobletNet. As shown in Figure 6, for GobletNet, the semantic encoding branch tends to focus on the overall objects, whereas the HF encoding branch focuses more on the texture and contour of images. In contrast, UNet cannot distinguish them well. Furthermore, compared with the encoder of UNet, the encoder of GobletNet has higher activation scores for segmentation objects (the activated objects are redder) and lower activation

TABLE XII
ABLATION OF VARIOUS COMPONENTS, INCLUDING WAVELET-BASED HF IMAGES, THE EXTRA CODING BRANCH OF HF IMAGES AND THE FAM

| HF Image | HF Branch | FAM | EPFL | | | | CREMI | | | | SNEMI3D | | | | UroCell | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ | Jaccard ↑ | Dice ↑ | 95HD ↓ | ASD ↓ |
| | | | 75.55 | 86.07 | 20.96 | 3.26 | 72.62 | 84.14 | 4.55 | 0.80 | 61.58 | 76.22 | 4.84 | 1.31 | 70.62 | 81.52 | 20.51 | 5.51 |
| ✓ | | | 76.05 | 86.40 | 23.25 | 3.47 | 79.31 | 88.46 | 3.67 | 0.63 | 63.39 | 77.59 | 4.77 | 1.26 | 73.15 | 83.54 | 18.71 | 5.13 |
| ✓ | ✓ | | 78.83 | 88.16 | 15.37 | 2.47 | 79.58 | 88.63 | 3.38 | 0.56 | 63.69 | 77.82 | 4.09 | 1.14 | 72.18 | 82.60 | 18.00 | 6.20 |
| ✓ | ✓ | ✓ | **81.66** | **89.91** | **12.04** | **2.12** | **80.17** | **89.00** | **3.02** | **0.54** | **64.18** | **78.18** | **3.58** | **1.04** | **74.42** | **84.61** | **13.47** | **4.45** |

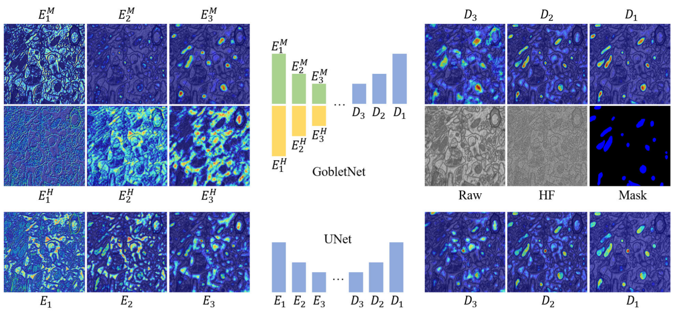


Fig. 6. Qualitative comparison of the class activation map of the encoders and decoders between UNet and GobletNet. We present the first three layers of encoders and the last three layers of decoders. For GobletNet, $E_i^M$ and $E_i^H$ indicate the $i$-th layers of the semantic encoder and HF encoder, respectively. For UNet, $E_i$ indicates the $i$-th layer of the encoder. $D_i$ indicates the $i$-th layer of the decoder.

scores for non-target regions (the non-activated regions are bluer).

## V. CONCLUSION

In this work, we quantitatively analyzed the high-frequency characteristics of EM images via the wavelet transform and identified **Characteristic 1** and **Characteristic 2**. To better utilize these characteristics, we propose a wavelet-based high-frequency image fusion network, GobletNet, which achieves state-of-the-art semantic segmentation of EM images. It was constructed based on encoder-decoder architecture without a complex design or complicated training process, which ensures practicality while achieving superior performance. The results of extensive experiments on seven public datasets demonstrate the effectiveness of our proposed model.

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